# STRUCTURAL ANALYSIS 1 (INS2512) - LESSON 3 -

Assist. Prof. Dr. Muzaffer BÖREKÇİ
Email: structuralanalysis1mb@gmail.com

# Assumptions in Structural Analysis

## 1) Material Behavior:

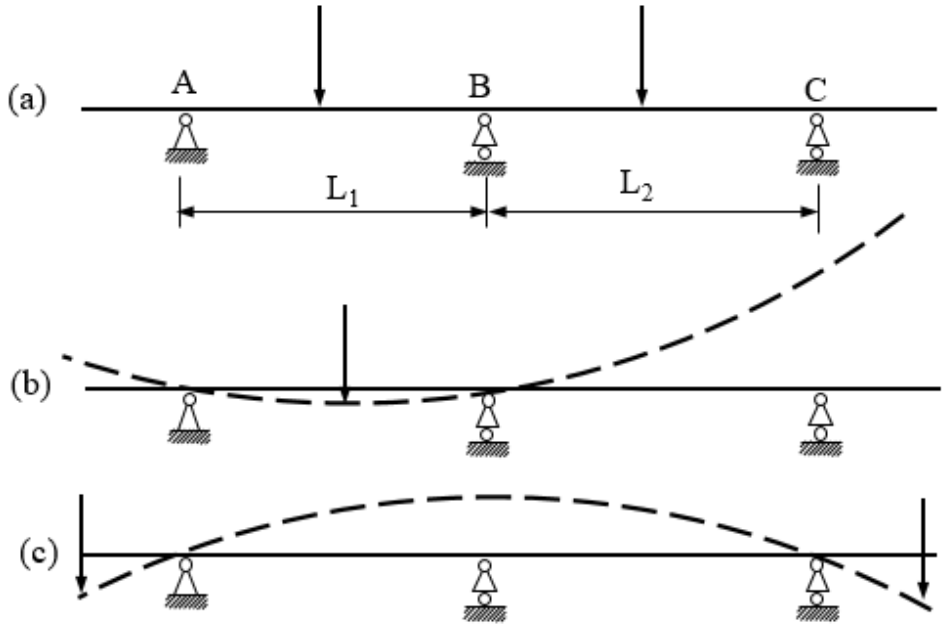
Most construction materials behave in accordance with Hooke's law up to certain limits.

In general, stresses under service loads are so low that the behavior is in the elastic range. In the context of this course, construction materials are assumed to be linear elastic, homogeneous and isotropic.

# Assumptions in Structural Analysis

## 2) Geometry:

The geometry of a structure considered in the context of structural analysis course is unchangeable, that is, it is independent of the magnitude and application of any load acting on the structure.



# Assumptions in Structural Analysis

## 3) Euler – Bernoulli Theory:

The Euler – Bernoulli theory is valid in the context of structural analysis course.

***Euler – Bernoulli Theory: Plane sections normal to the neutral axis before bending remain plane and normal to the neutral axis after bending.***

# Assumptions in Structural Analysis

4) Displacements:

Displacements due to external loads are so small compared to the length of structural members that they can be considered as negligible and it can be assumed that the original geometry of the structure remains unchanged.

Thus, equilibrium equations can be written for the undeformed structure. This is referred to as first-order theory.

If the equilibrium equations are written for the deformed structure, it is referred to as second-order theory.

**In the context of structural analysis course, first-order theory is valid.**

# Assumptions in Structural Analysis

5) Superposition Principle:

Superposition principle is valid in case of material is linear elastic and first-order theory is valid.

According to this principle, *the total displacement or internal loadings (stress) at a point in a structure subjected to several external loadings can be determined by adding together the displacements or internal loadings (stress) caused by each of the external loads acting separately*.

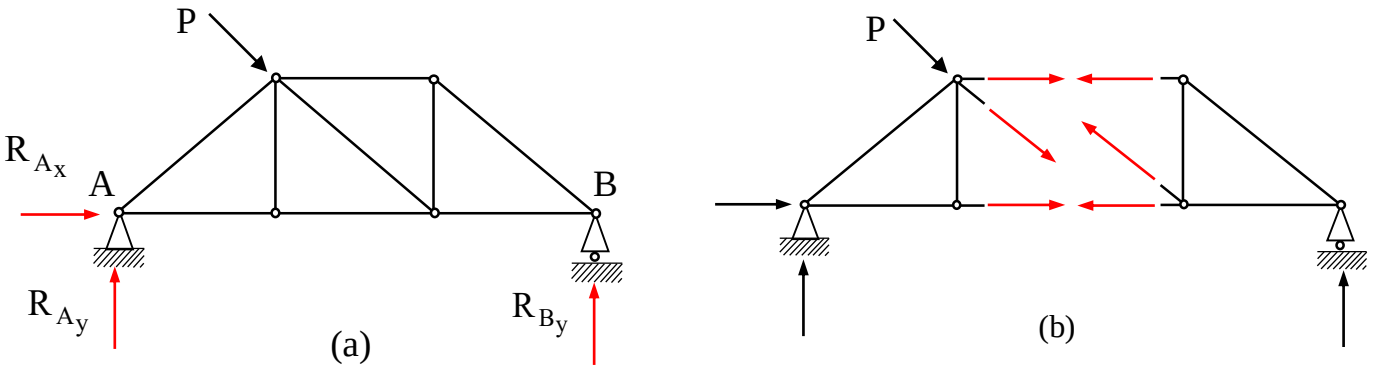
6) Stability:

All structures considered in the context of this course are statically stable. Statically unstable structures cannot be examined in this course.

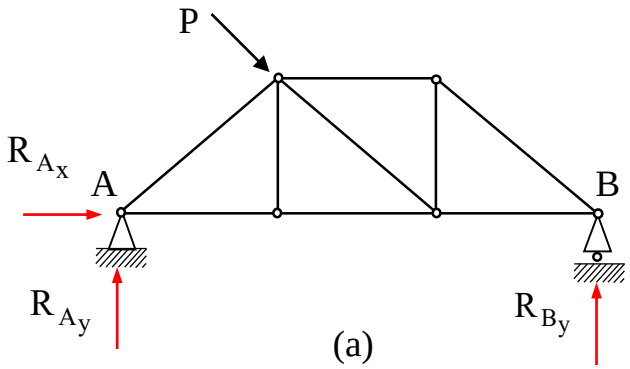
# STRUCTURAL SYSTEMS

## 1. Free Body

Free body can be whole system or a part of the structural system detached properly from the rest of the system under the effect of whole loads. It is important to be able to draw free body diagram accurately for solving a structural analysis problem.

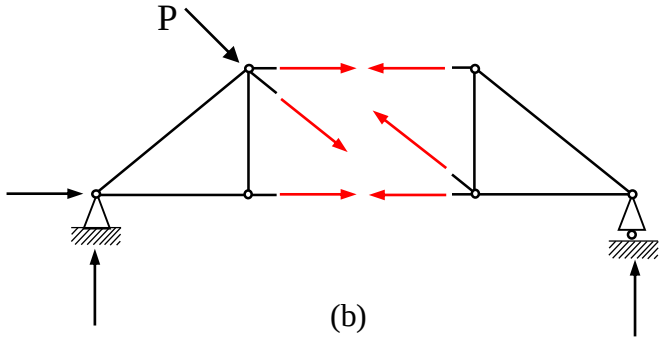


Consider a truss structure shown in the figure:



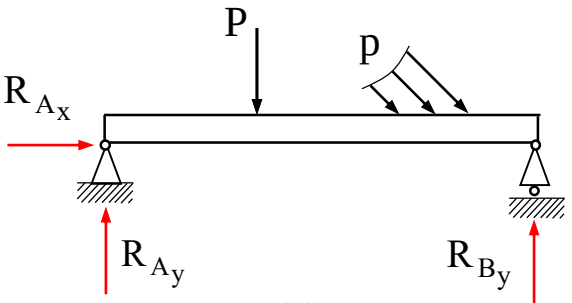

The free body diagram of the whole structure under the external load P is shown in the figure. Since force of truss elements are the internal forces of the free body, they are not seen in the figure. The equilibrium of this free body gives support reactions.

Let’s split the structure into two and draw the free body diagram:




Each part of the structure is a free body. Internal forces in the cut elements are considered as external loads. A structure must be split into parts in such a way that internal forces of interest act on the free body as external loads. The equilibrium conditions of these free bodies gives frame forces of cut frame of truss.

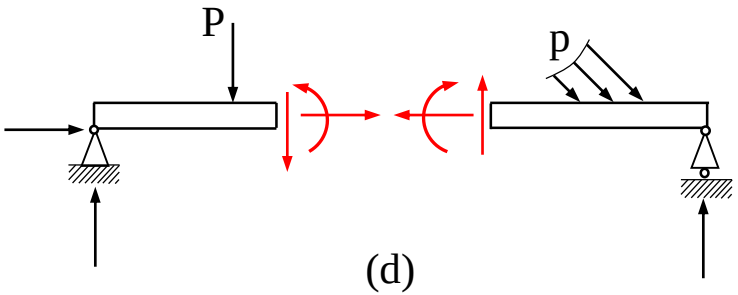
Consider a simply supported beam under external loads. The free body diagram of the beam is given below:




(c)

Given free body is for whole system. The support reactions can be determined from the free body diagram of the whole structure.

Then, the beam can be splitted into two parts in order to determine internal loads at a section of interest.




(d)

The internal forces acting on the cut section are now considered as external loads and they can be determined from the equilibrium conditions of the free body.

# 2. Loads

There are three types of loads affecting the structural systems:

- External Loads
- Support Reactions
- Internal Forces

External Loads:

Detailed information about external loads has been given in previous lessons.

Support Reactions:

Support reactions are the reaction loads formed at the supports as a result of external loads. Support reactions should be estimated before the estimation of internal loads.

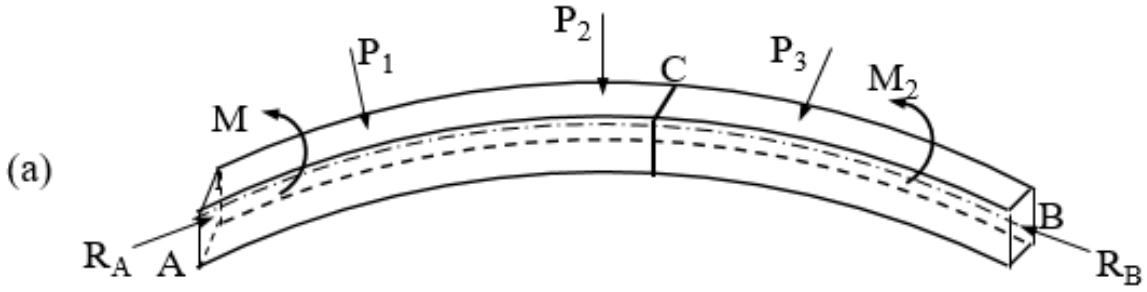
## Internal Forces:

An internal force acting on a rigid body causes stresses within the body. The stresses occur not only at application point of force but also at the other points of the body since the resulting stresses are tranferred to the other points by molecules as a result of action-reaction principle.

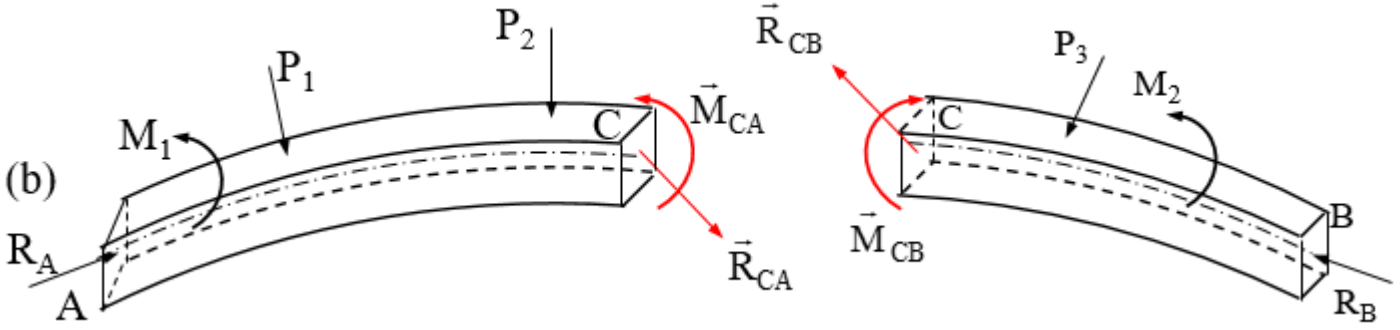
The resultant of stresses are transferred to as internal forces.

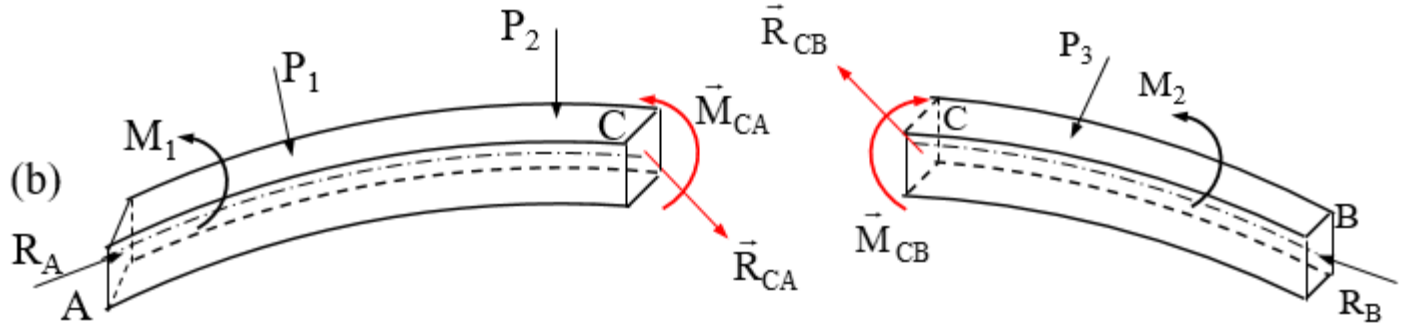
# Internal forces of a 1D element in a plane:

Consider a 1D structural element under external loads.




The element is in equilibrium. Let's cut this element into two at C.





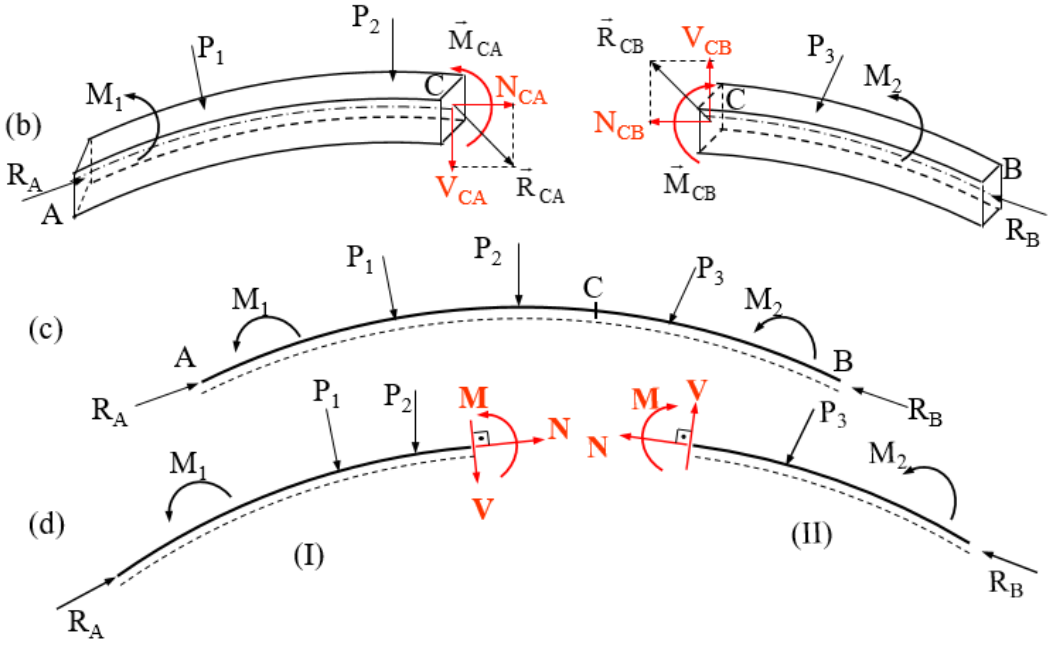

The loads applied by the elements AC and CB to each other can be reduced to a force and moment vector $\vec{R}$ and $\vec{M}$ within the same plane. The applied loads are equal to each other according to the action-reaction principle.

$$\vec{\boldsymbol{R}}_{\boldsymbol{CA}} = -\vec{\boldsymbol{R}}_{\boldsymbol{CB}} \qquad \vec{\boldsymbol{M}}_{\boldsymbol{CA}} = -\vec{\boldsymbol{M}}_{\boldsymbol{CB}}$$

The component of $\vec{R}$ parallel to the axis of element is referred to as axial force (N).

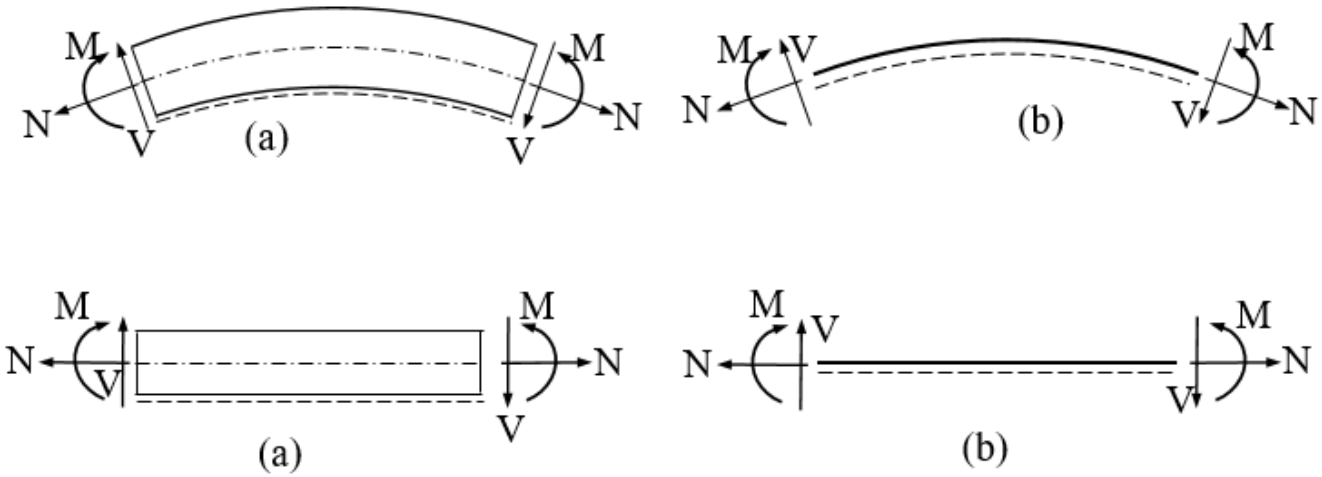
The component of $\vec{R}$ perpendicular to the axis of element is referred to as shear force (V).

The moment $\vec{M}$ is referred to as bending moment (M).




There exist **three** internal forces acting on a section of 1D structural element in a plane.

Positive sign convention of internal forces are given below:

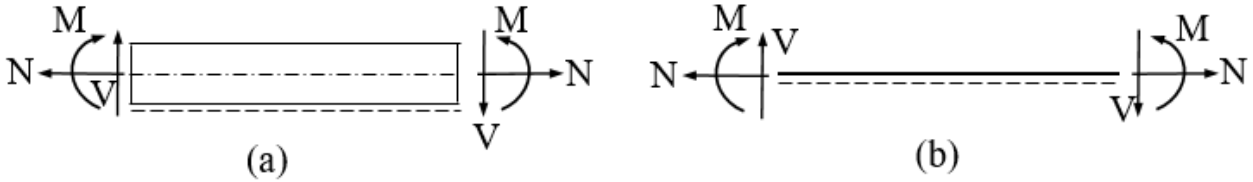



While presenting the sign convention, one side of the element is marked with a dashed line

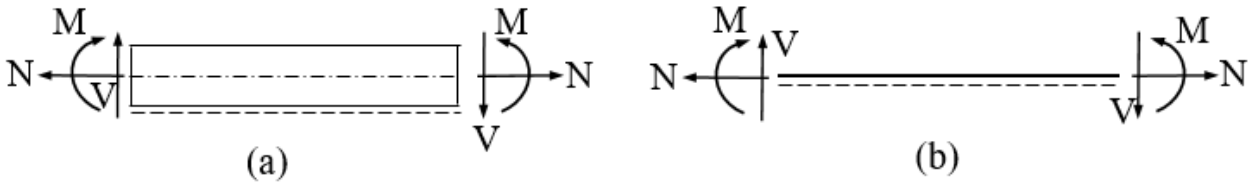
Axial Force: It is denoted with N. Normal force is the component of $\vec{R}$ parallel to the axis of structural element. It is also referred to as axial load.

If it causes tension, the sign is positive (+), if it causes compression, then it is negative (-).

Shear Force: It is denoted with V (in some references Q or T). It is the component of $\vec{R}$ perpendicular to the axis of structural element. If shear force tends to rotate the element in clockwise direction, the sign is positive (+). Otherwise, it is negative (-).

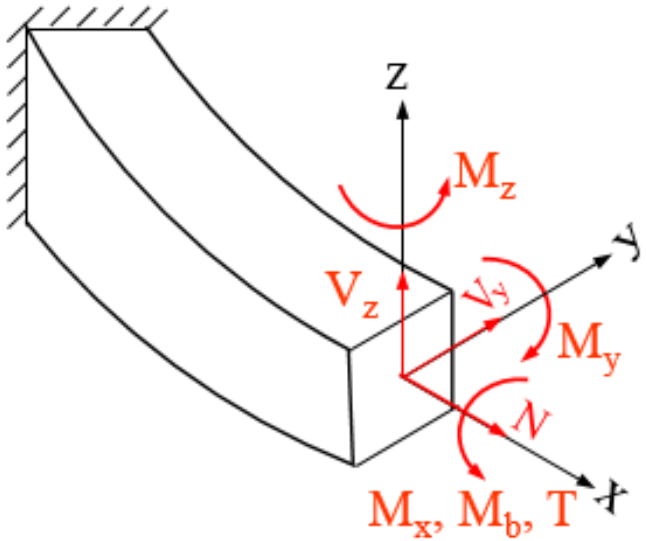


Bending Moment: It is denoted with M. The moment trying to elongate the marked side of the element is positive (+), otherwise it is negative (-).



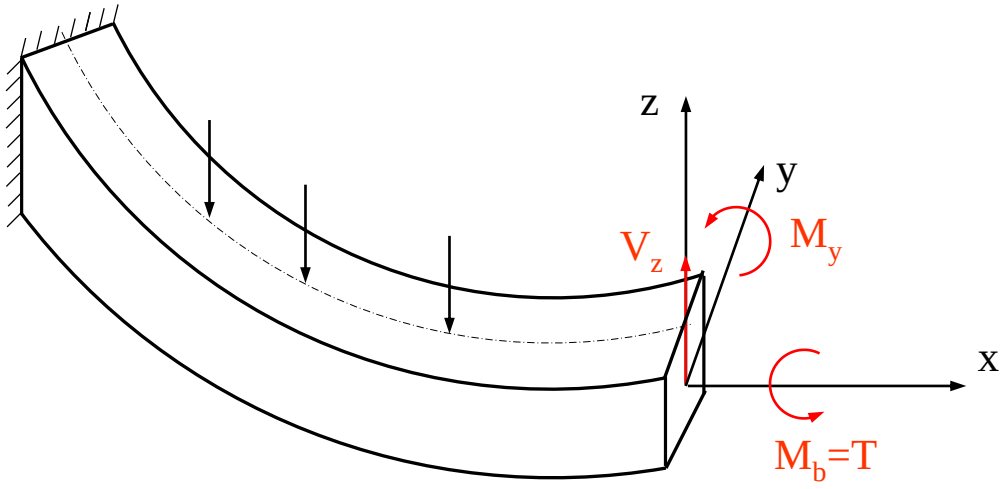
# Internal forces in an element in 3D:

At a section of a structural element in 3D, there exist 6 internal loads, three of them are moments, two of them are shear force and one of them is axial force.



## Internal forcess in a curved element subjected to external loads normal to the plane of element:

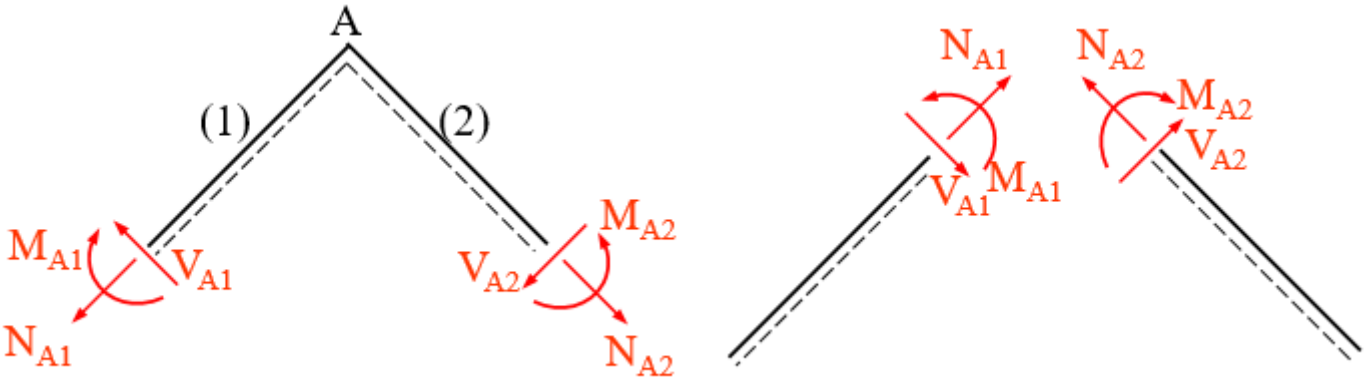
There exist three internal forces in such an element:




- Shear load parallel to external load (Vz)
- Torsional moment
- Bending moment My respect to the axis perpendicular to the external load

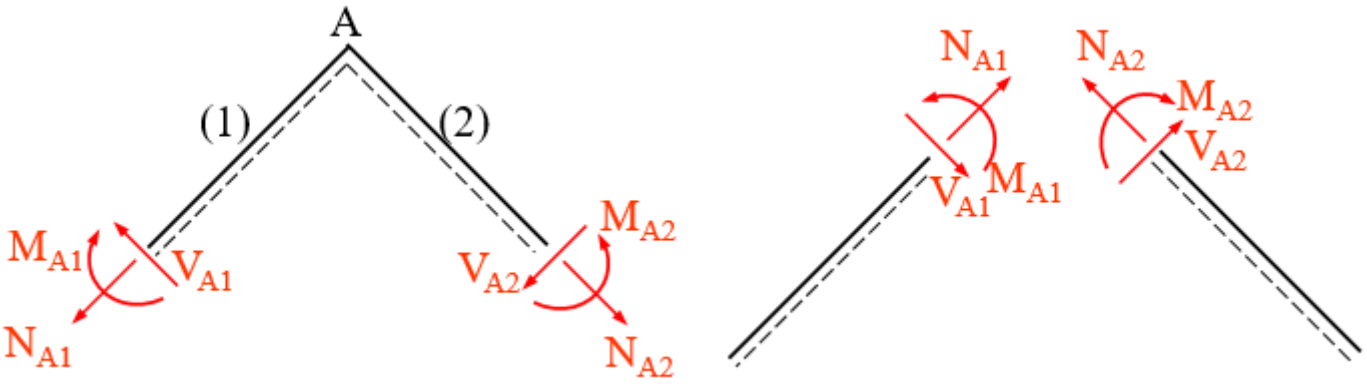
# Internal loads at the joints of planer systems:

Fixed (Rigid) Joint: There exist three unknown internal forces in each structural element framing into the same fixed joint (axial force, shear force and bending moment).



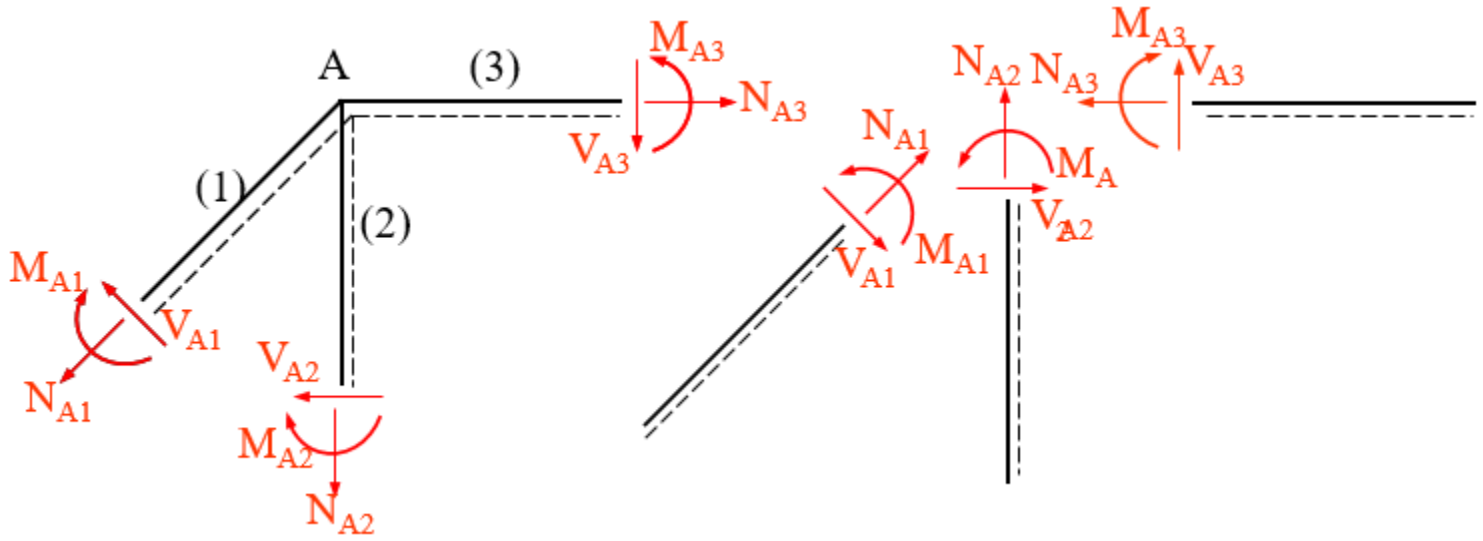

If the internal forces in an element is known in case of a fixed joint at which two elements are connected, then the internal forces in the other element can be determined by means of the action-reaction principle.

If the internal forces in an element is known in case of a fixed joint at which two elements are connected, then the internal forces in the other element can be determined by means of the action-reaction principle.




Hence, the number of unknowns at a fixed joint at which two elements are connected is three.

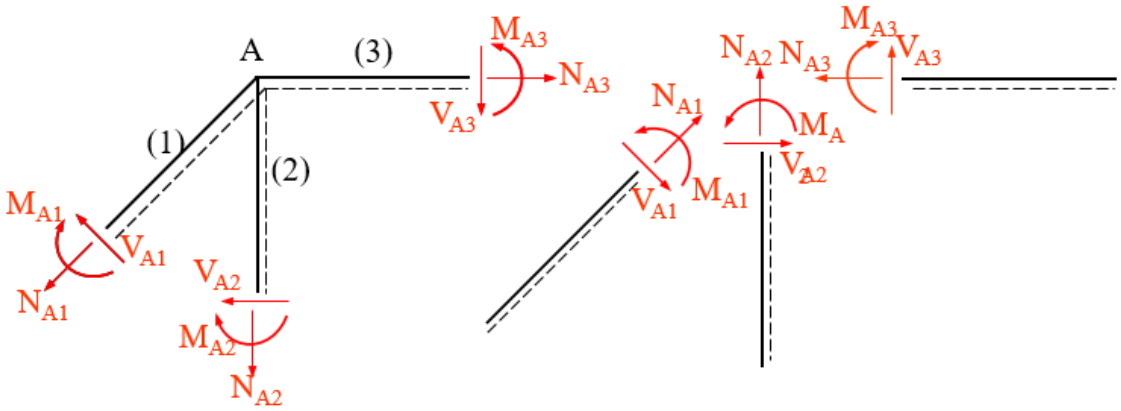
In case of a fixed joint at which three elements are connected, if the internal forces in two elements are known, then the internal forces in the third element can be determined.



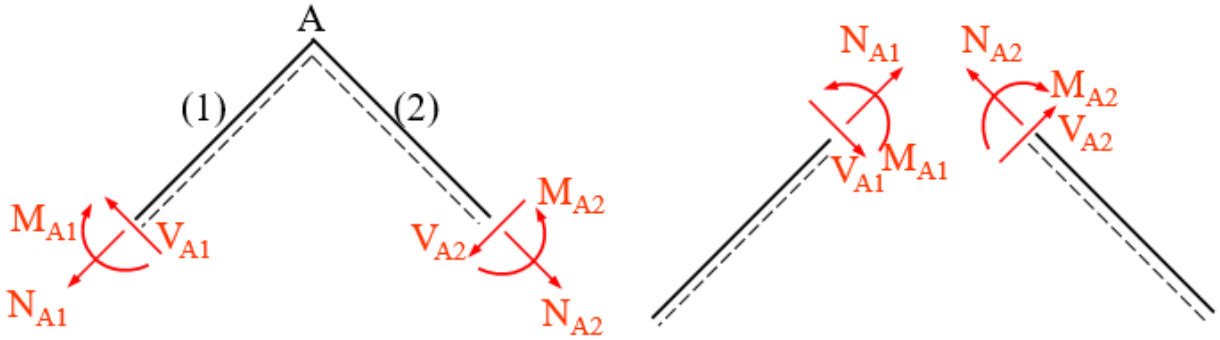

In other words, the number of unknowns at a fixed joint at which three elements are connected is six.

Therefore, the number of unknown (n) at a fixed joint at which $n_{el}$ elements are connected is:

$$n = 3(n_{el} - 1)$$

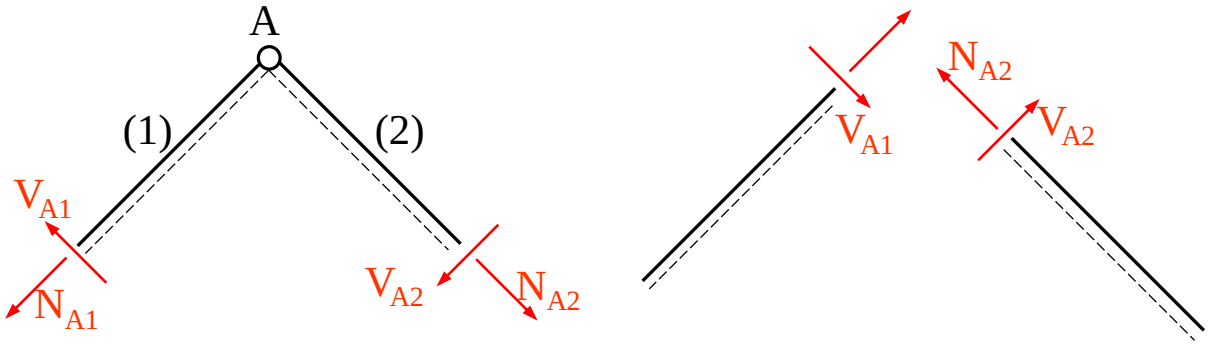



**n = 3*(3-1) = 6**



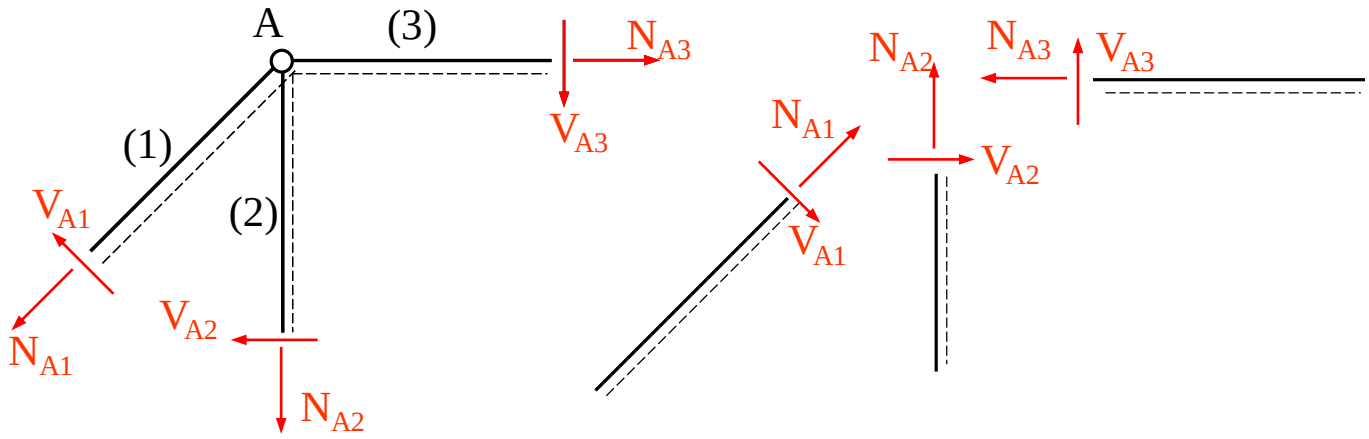

**n = 3*(2-1) = 3**

Pin Joints: At the end of structural elements framing into pin joints, there exist axial and shear forces but moments do not occur at these ends since relative rotation of elements are not restrained.



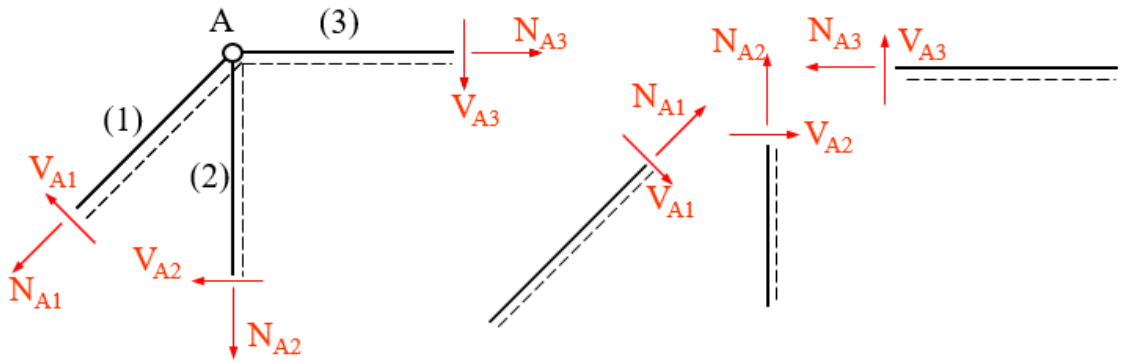

Hence, the number of unknowns at a pin joint at which two elements are connected is two.

The number of unknowns at a pin joint at which three elements are connected is four.

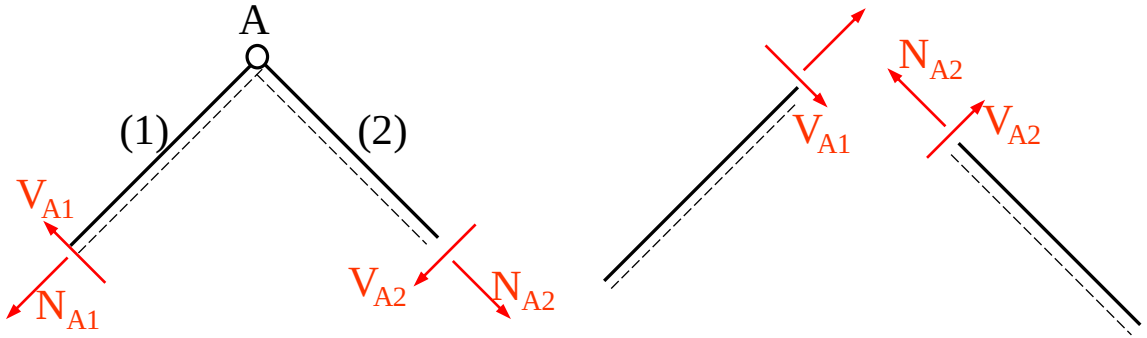


Therefore, the number of unknown (n) at a pin joint at which $n_{el}$ elements are connected is:

$$\boldsymbol{n = 2(n_{el} - 1)}$$




**n = 2*(3-1) = 4**




**n = 2*(2-1) = 2**

There are also axial hinge and shear hinge joints.



Axial hinge



Shear hinge

At an axial hinge joint, the unknowns are shear force and bending moment.

At a shear hinge joint, the unknowns are axial force and bending moment.

# 3. Equations of Equilibrium

You can recall from statics course that a structure or one of its elements is in equilibrium when it maintains a balance of load and moment.

In general, this required that the load and moment equations of equilibrium be satisfied along three independent axes.

$$\sum P_x = 0 \qquad \sum M_x = 0$$

$$\sum P_y = 0 \qquad \sum M_y = 0$$

$$\sum P_z = 0 \qquad \sum M_z = 0$$

The principal load-carrying portions of most structures lie in a single plane and the loads are also coplanar. Then the requirements for equilibrium reduce to

$$\sum P_x = 0 \qquad \sum P_y = 0 \qquad \sum M_0 = 0$$

$\sum P_x$ and $\sum P_y$ represent the algebraic sums of the x and y components of all the loads acting on the structure or a part of the structure and $\sum M_0$ represents the algebraic sum of the moments of these load components about an axis perpendicular to x-y plane (z axis) and passing through the point O.